IoT Based Industrial Air Quality Monitoring System

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***Abstract*— In the past two decades, researchers have accelerated a number of different methods to monitor and reduce air contamination leading to the development of efficient and effective air quality measuring systems using air purifiers. Although, recent technological advancements such as IoT (Internet of Things) with Cloud Computing have allowed researchers to obtain and monitor real-time data. In this report, an IoT-enabled industrial air quality monitoring device is proposed. This device is enabled with an MQ-135 gas sensor for precisely monitoring the air quality and detecting the presence of foreign contaminants such as alcohol. The proposed device also uses a Node MCU ESP 8266 Wi-Fi module to efficiently transmit real time data to a smart device (E.g. Smartphone) using an IoT platform.**

Keywords*— *IoT, Air Quality Index (AQI), Air Pollution, MQ-135, Node MCU ESP 8266

I. INTRODUCTION

Today, air pollution is a major environmental concern. Atmospheric circumstances endure to weaken early causing devastating effects on climate and the ecosystem. Many solemn health issues as respiratory problems, cardiovascular issues and dermatological diseases in humans are caused by air pollution [1][2]. The air is polluted by toxic gases and contaminants. The pollutants released from various industrial plants and factories, vehicle emissions and countless other sources consequently increase the concentration of greenhouse gases as CO_2, NO_2 and SPM (Suspended Particulate Matter) in the atmosphere. Now more than ever, the need for monitoring air quality is of utmost importance. The production of affordable IoT-based air quality monitoring devices and their widespread use have allowed people to detect the air quality in concerned areas swiftly [3][4]. In this article, one real-time portable quality of air monitoring system based on several air parameters like the level of CO_2, the presence of smoke, alcohol and other foreign contaminants is proposed.

II. AIR QUALITY PARAMETERS

To regulate the air quality from the data obtained, the propounded IoT-based industrial air quality monitoring system uses pre-defined standards for indoor air quality [Table 1] through levels of CO_2 in parts per million (ppm) as mentioned in Indoor Air Quality Control Act [5][6]

TABLE I. STANDARDS FOR CO_2 LEVELS OF INDOOR AIR

CO_2 Concentration	Health effects
< 400 ppm	Normal outdoor level- no observable effect [6].
400-2000 ppm	Respiratory problems, severe headaches, loss of cognitive function, eyesight impairment, dizziness are observed [6].
2000-5000 ppm	Increased heart rate, restrictive lung behavior, unhealthy blood carbon dioxide levels, drowsiness and nausea are observed [6].
5000-40,000 ppm	Sleep disruption, lethargy, mental slowness and increased lung dead space volume are observed [6].
>40,000 ppm	Exposure to this level will lead to severe oxygen deprivation which will cause lifelong brain damage, paralysis, coma or may even prove to be fatal [6].

These thresholds may vary based on area or location. Hence, the device needs to be calibrated accordingly.

III. IOT PLATFORM FOR AIR QUALITY MONITORING

In the proposed device, a Node MCU ESP8266 12E (Wi-Fi enabled) board is used and it is programmed using the Arduino IDE (Integrated Development Environment) with an ESP 8266 add-on. Furthermore, to compile the program, a few additional libraries such as the Adafruit Unified Sensor library, have been installed. A standalone application named Blynk is used in order to provide a platform to connect the Node MCU ESP 8266 board over Wi-Fi. This allows users with specified access to remotely monitor the air quality of a place, as detected by the sensor, in real time via their smartphones [7][8].

ALGORITHM:-

1. START
2. Include libraries ESP8266WiFi.h, BlynkSimpleEsp8266.h : including the required libraries
3. Input character array auth[] : to store authentication token from Blynk application
4. Input character array ssid[] : to store the WiFi ID number
5. Input character array pass[] : to store WiFi password
6. Input integer n : declaring integer n
7. Start function Void Setup() : function to start the WiFi module and Blynk system

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> Invoke function Serial begin() : function to start WiF module
> Invoke function Blynk begin(auth, ssid, pass) : function to start the Blynk platform
> Invoke function pinMode(16, OUTPUT) : initiating the 16 number pin of Esp8266 module
> Invoke function pinMode(5,OUTPUT) : initiating the 5 number pin of Esp8266 module
> Invoke function timer.setInterval(1000L, sendUptime) : Initializing the delay time between the obtained data and the total uptime of the system

8. Start function Void sendUptime() : function to send uptime
> Invoke function Blynk virtualWrite(V1, n) : function to send uptime data to blynk app

9. Start function Void loop()
> Invoke function Blynkrun()
> Invoke function Timerrun()
> Initializing the value of integer n : feeding the data from Mq135 sensor to integer n
> Printing the value of n : printing the data from the sensor to the blynk application
> Check if the value of n is greater than 630 : If the condition is satisfied then pin number 16 will be HIGH and pin number 5 will be LOW after a delay of 2 seconds
> Invoke function digitalWrite(16, HIGH)
> Invoke function digitalWrite(5, LOW)
> Invoke function delay(2000)
> Check if the value of n is lesser than 630 : If the condition is satisfied then pin number 16 will be LOW and pin number 5 will be HIGH after a delay of 2 seconds
> Invoke function digitalWrite(16, LOW)
> Invoke function digitalWrite(5, HIGH)
> Invoke function delay(2000)

10. END

IV. HARWARE DESIGN

A. Air Quality Monitoring Device

The proposed device consists of an MQ-135 sensor which can detect the level of gases such as CO_2 and the presence of pollutants like Ammonia (NH_3), alcohol or smoke. The above-mentioned sensor has a digital and an analog output pin. When the levels of these gases go beyond a particular limit in the air, the output of the digital pin gets high and the analog output pin gives a voltage which can be used to approximate the level of such gases in the air [9][10]. Apart from the MQ-135 gas sensor and the Node MCU ESP8266 Wi-Fi board, the device also has an LED bulb that functions as a warning light, turning ON when the air quality is deemed unsatisfactory. The practical images of the device are shown in Fig. 1 and Fig. 2.

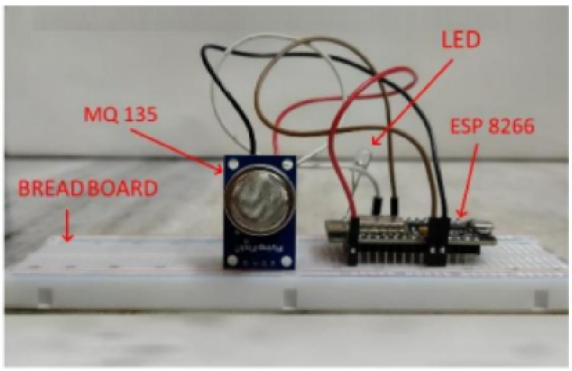


Fig. 1. Air Quality Monitoring Device (Front View)

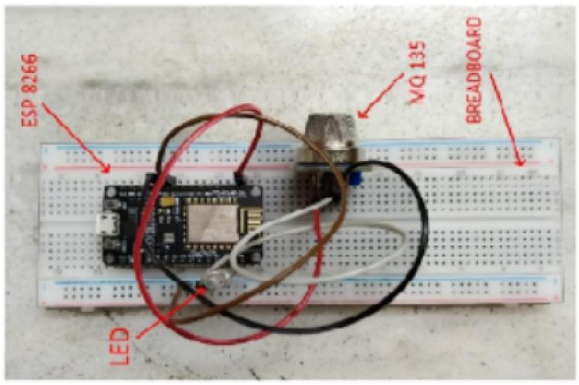


Fig. 2. Air Quality Monitoring Device (Top View)

B. A Remote Controlled Robot for additional mobility

In order to enhance the mobility of our system, it is fixed on a remote controlled robot. The robot uses a custom made cardboard chassis, an L298N motor driver, two standard issue BO motors and wheels connected to a series of 9V batteries (12V Lithium Ion batteries can also be used). The motor driver is controlled using a 433 MHz Wireless Radio Frequency Transmitter Receiver Board which has HT12D HT12E (decoder, encoder respectively). The motor driver can operate at up to 35 Volt DC and is able to control maximu 4 DC motors (without direction and speed control) or 2 DC motors (with direction and speed control). The transmitter-receiver module has four switches. These switches controls the forward and backward rotation of each motor and under ideal conditions, the module should have a range of a 100 Meters. The primary usage of this robot is to carry the above-mentioned Air Quality Monitoring device to areas where the air quality is required to be measured (especially if said area is hard to reach manually or if it is suspected to be hazardous for humans). The prototype image of the remote-controlled robot is shown in Fig. 3.

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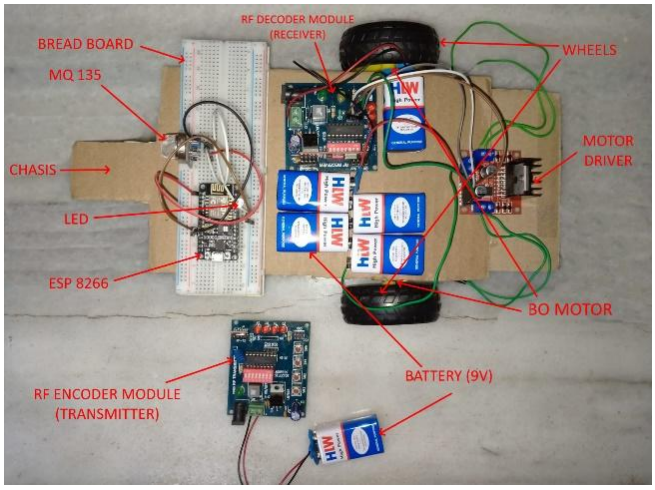


Fig. 3. Remote Controlled Robot carrying the Air Quality Monitoring System

The block diagram given below in Fig. 4 shows the working of the designed monitoring system for Air Quality device and the Remote Controlled Robot which together form our system.

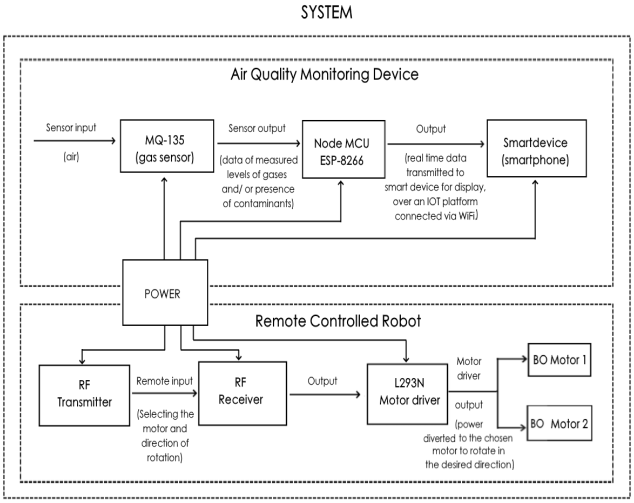
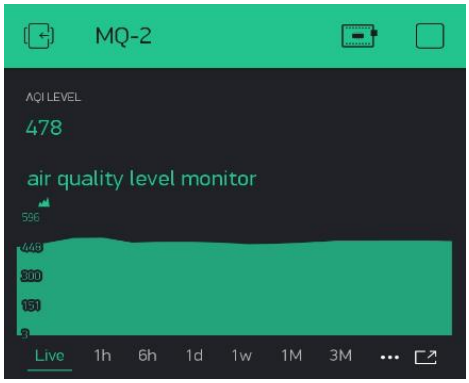


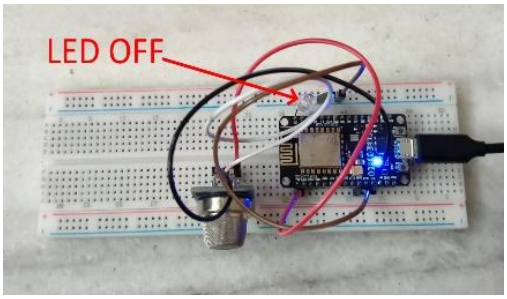
Fig. 4. Block Diagram of the Monitoring of Air Quality Device and the Remote Controlled Robot

V. Output

The Air Quality Index (AQI) reading in ppm as seen via the user interface of the Blynk IoT platform and the state of the proposed system under normal conditions is shown in Fig. 5(a) and Fig. 5(b) respectively, whereas the AQI reading in ppm as seen via the user interface of the Blynk IoT platform and the state of the proposed system under hazardous conditions is publicized in Fig. 6(a) and Fig. 6(b) respectively.



(a)

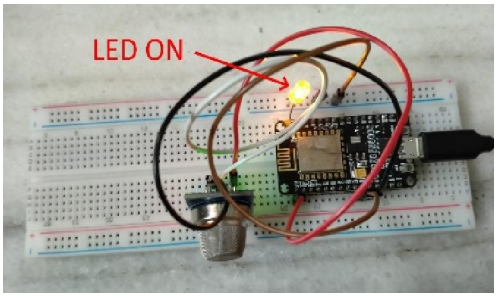


(b)

Fig. 5 (a) AQI reading as seen in real time via mobile interface of Blynk IoT platform under normal conditions
(b) Air Quality Monitoring System under normal conditions



(a)



(b)

Fig. 6 (a) AQI reading as seen in real time via mobile interface of Blynk IoT platform under hazardous conditions
(b) Air Quality Monitoring System under hazardous conditions

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VI. Conclusion

The proposed device allows the air quality data which is measured by the MQ-135 sensor to be observed via the ESP8266 Wi-Fi module and the user interface of the Blynk IoT platform in real time. From Fig. 5(a), the Air Quality Index is observed to be 478 ppm [Table 1] which close to the acceptable range while in Fig. 6(a), the Air Quality Index is observed to be 606 ppm [Table 1] and is significantly larger than the acceptable levels[11][12]. This device is unique in its inclusion of a remote controlled robot utilizing a 433 MHz Wireless RF Transmitter Receiver Board which carries the device itself (as shown in Fig.3) allowing easy access to difficult or hazardous areas. It is also worth mentioning that this device has a fairly simple structure and an editable program allowing users to easily modify the threshold levels according to the environment in which it is to be used.

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References

[1] United States Environmental Protection Agency, *Managing air quality - air pollutant types*, October 2018,

[2] S. Bhattacharya, S. Sridevi, and R. Pitchiah, "Indoor air quality monitoring using wireless sensor network," in *2012 Sixth International Conference on Sensing Technology (ICST)*, pp. 422–427, Kolkata, India, December 2012.

[3] G. Parmar, S. Lakhani, and M. Chattopadhyay, "An IoT based low cost air pollution monitoring system," in *2017 International Conference on Recent Innovations in Signal processing and Embedded Systems (RISE)*, Bhopal, India, October 2017.

[4] M. Tastan and H. Gokozan, "Real-time monitoring of indoor air quality with internet of things-based E-nose.," *Applied Sciences*, vol. 9, no. 16, article 3435, 2019.

[5] S. Devarakonda, P. Sevusu, H. Liu, R. Liu, L. Iftode, and B. Nath, "Real-time air quality monitoring through mobile sensing in metropolitan areas," in *UrbComp '13 Proceedings of the 2nd ACM SIGKDD International Workshop on Urban Computing*, Chicago, Illinois, August 2013.

[6] Bamrungwong, N.; Vongmanee, V.; Rattanawong, W. The Development of a CO_2 Emission Coefficient for Medium- and Heavy-Duty Vehicles with Different Road Slope Conditions Using Multiple Linear Regression, and Considering the Health Effects. *Sustainability* 2020, *12*, 6994.

[7] H. Ghayvat, S. Mukhopadhyay, X. Gui, and N. Suryadevara, "WSN- and IoT-based smart homes and their extension to smart buildings," *Sensors*, vol. 15, no. 5, pp. 10350–10379, 2015.

[8] M. Benammar, A. Abdaoui, S. Ahmad, F. Touati, and A. Kadri, "A modular IoT platform for real-time indoor air quality monitoring," *Sensors*, vol. 18, no. 2, p. 581, 2018.

[9] K. Gupta and N. Rakesh, "IoT based automobile air pollution monitoring system," in *2018 8th International Conference on Cloud Computing, Data Science & Engineering (Confluence)*, Noida, India, January 2018.

[10] A. Rackes, T. Ben-David, and M. S. Waring, "Sensor networks for routine indoor air quality monitoring in buildings: impacts of placement, accuracy, and number of sensors," *Science and Technology for the Built Environment*, vol. 24, no. 2, pp. 188–197, 2018.

[11] G. Rout, S. Karuturi, and T. N. Padmini, "Pollution monitoring system using IoT," *ARPN Journal of Engineering and Applied Sciences*, vol. 13, pp. 2116–2123, 2018.

[12] K. A. Kulkarni and M. S. Zambare, "The impact study of houseplants in purification of environment using wireless sensor network," *Wireless Sensor Network*, vol. 10, no. 03, pp. 59–69, 2018.

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